European Developer Conference



Palm OS 6 “Sneak Peek”

Presented By:
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Director,
Developer
Architecture
and Disclosure



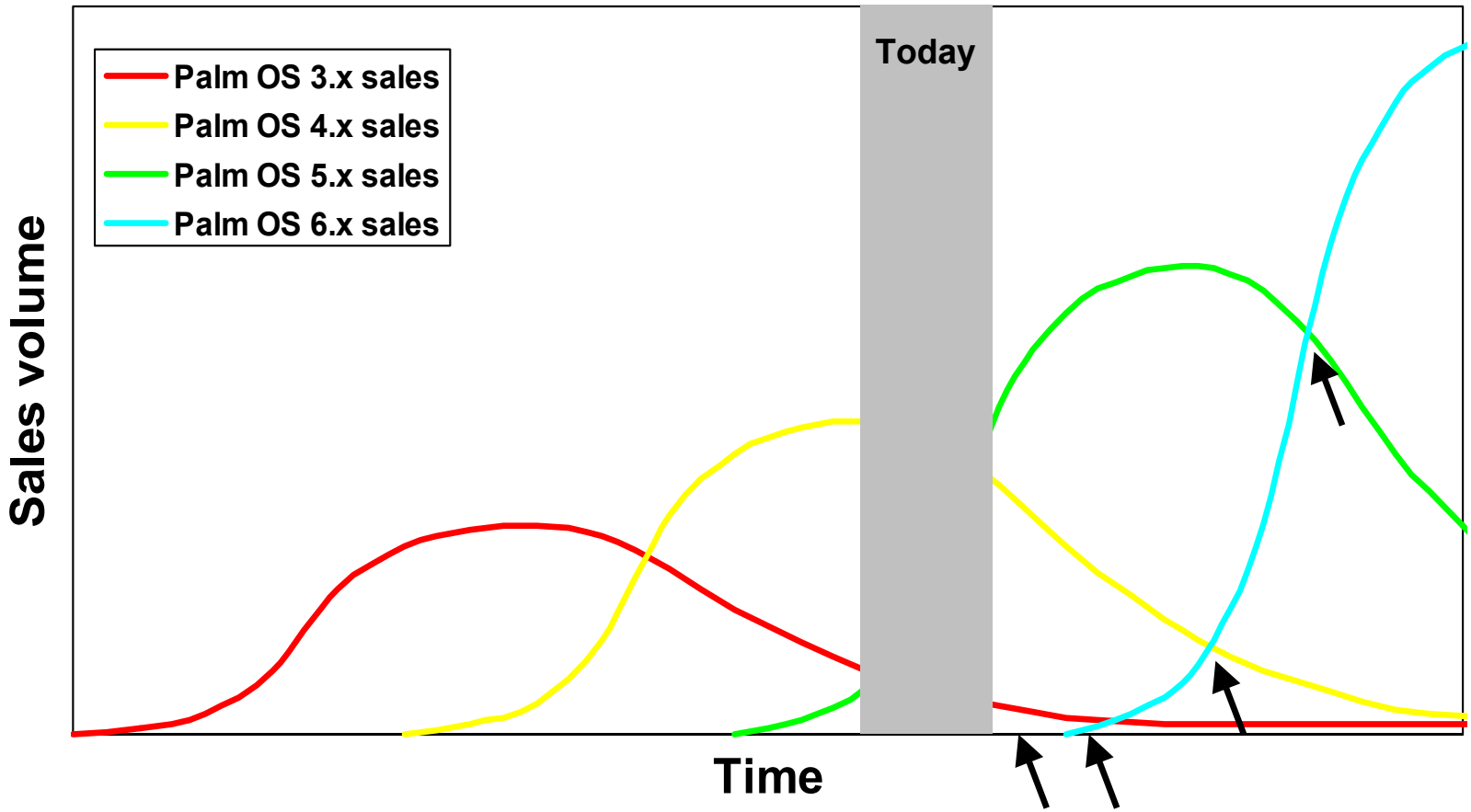
Why this talk?

- You need to plan your future
- PalmSource is creating Palm OS 6
- Talking about the core technology, as opposed to the applications thereof
- Time to prepare

(Disclaimer: forward-looking statements abound, but the future cannot be guaranteed!)

Idealized Palm OS sales

(Chart is for illustration purposes only; shows neither actual nor projected sales)



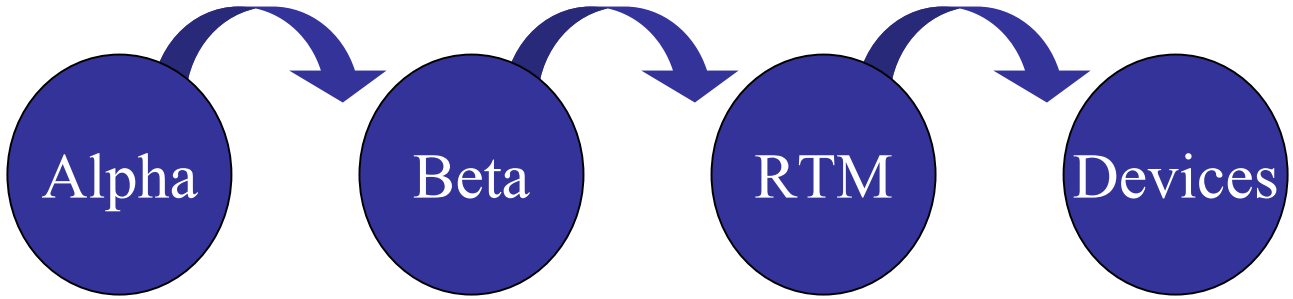
Three effect categories

Every new OS release affects you in 3 ways:

- Existing application compatibility
  - (users upgrading their device)
- New device and OS features
  - (with your old application)
- New APIs, new tools, new SDKs
  - (for your new app releases)

Timeframes

- Palm OS 6 is in its Beta cycle
- It is scheduled for handoff to licensees in the fall
- Timing of new devices is up to Palm OS licensees



Existing App Compatibility

Compatibility is similar to Palm OS 5.x situation

- PACE runs your existing 68k PRCs
- PNOs in your 68k PRCs still ideal for speed optimization

Assumptions about OS implementation are dangerous

- The internals are considerably revamped

High degrees of compatibility are expected

- (Assuming your applications are well behaved!)

Change is threatening

People want big improvements.

People want nothing to change.

So… nothing is changing,

and everything is changing.

Changing everything, but nothing

For PACE apps, the world doesn't change

- (much.)
- This is a perfectly valid model to use for years.

New APIs and capabilities are available to native applications

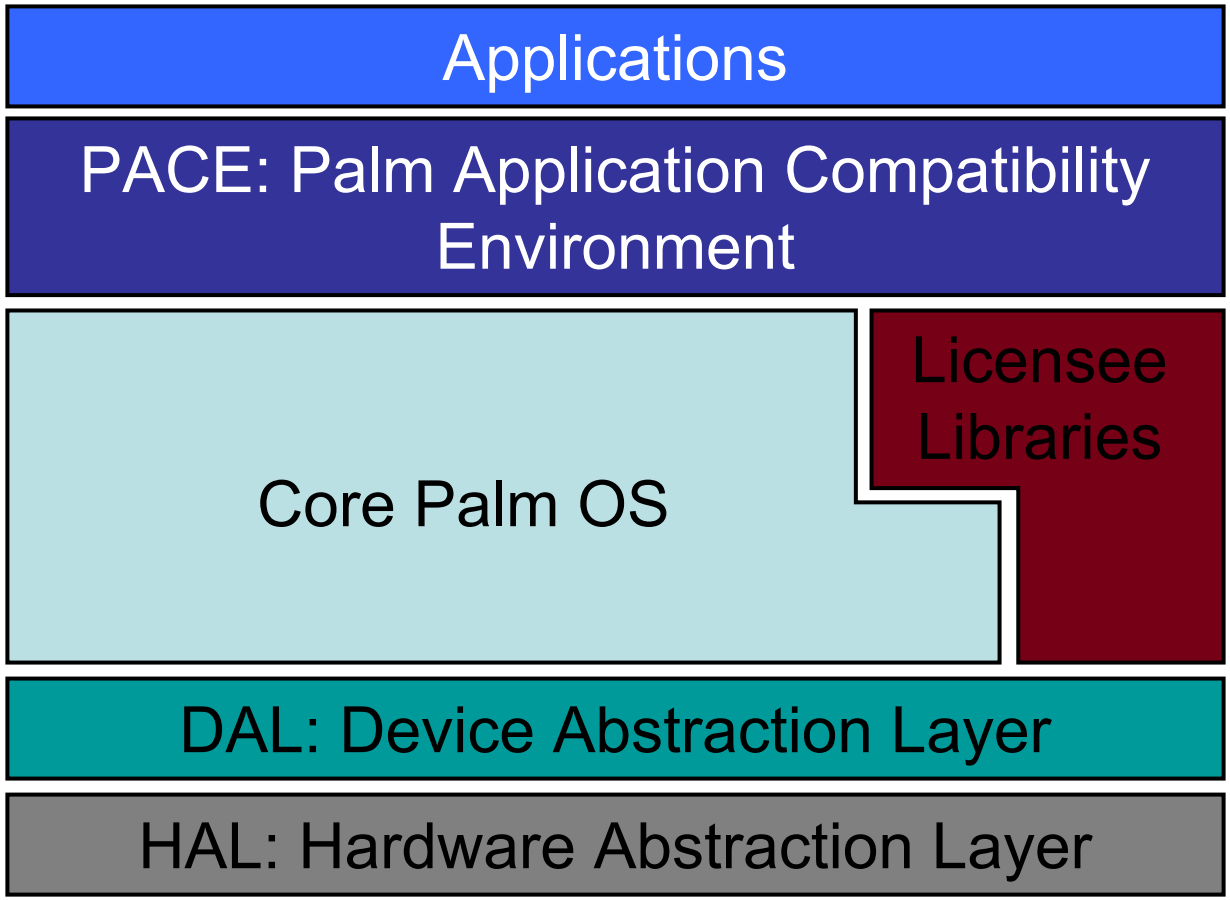
- This does involve writing new code and tweaking old code.
- Native ARM applications require Palm OS 6.

Palm OS 6 Developer Features

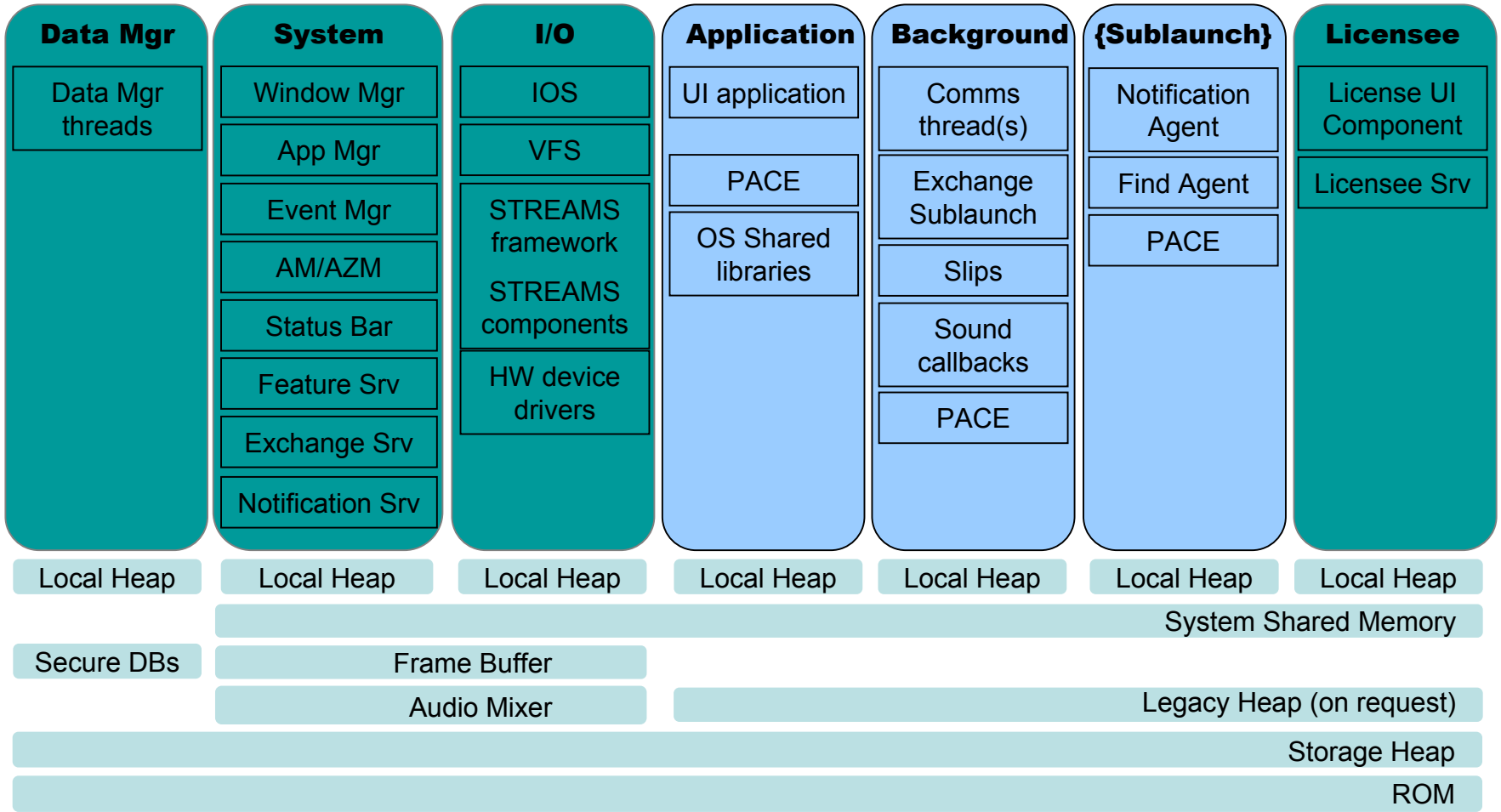
- **Protected memory system**
- **Process protection**
  - (stability and security)
- **New runtime model for flexible apps & libs**
- **Extended database architecture**
  - (interoperability and extensibility)
- **Threading and Background processes**
- **New Communications architecture**

… and many more

Block Diagram for 5.x



Palm OS 6.0 Processes and Memory Mapping



Removing limitations

The top 3 old limitations: gone!

- No 64k limit on records, resources or code segments
- No global variable restrictions
- No 15 category limit

Many new APIs

- Details will be provided

Benefits are ready when your app is ready

New model implications

Some things change for your existing apps

- Not all events go to your event loop
- The hack-like notifications, patching, etc.
- Code signing & patching

Unless you're extending the system, these shouldn't be noticed by your application

What will be different

- **The display**
  - Screen sizes, dimensions, and active input areas
  - Direct access to screen buffer not allowed
    - For security reasons, and for new features e.g. acceleration
    - Applications have back buffering for compatibility
- **Memory model: new capabilities**
- **Schema databases available**
- **Endianness, structure packing…**

The Tools

Existing tools, sources and SDKS

- Your compiler, POSE, PalmSim 5.x
- Still work just like before.

New tools and SDKS

- New compiler, debugger, Emulator, PalmSim
- Sources need to handle endianness, and changes typical of an SDK and OS update, when you choose.
- New IDE integrations

Mixing the Old and the New

Mixing old and new data

- Design wire formats
- Consider the endianness of records & resource data in PDBs
- If you change your DBs, change your conduits

Mixing the Old and the New (cont)

Mixing old and new code

- 68k app with PNO code resources
- 68k apps calling ARM shared libraries, sublaunching, etc.
- Can you, and should you, create a “fat” prc

Deciding if and when to go native

- PACE today, PACE tomorrow…

More to come

This is just a sneak peek of the low levels

Tons of new APIs and capabilities, leading to new categories of solutions

Today's Palm OS applications have a great future

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Q & A